Counting Polygon Free Mosaics

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Abstract

Hong and Oh introduced a model for multiple ring polymers in physics in which 7 distinct tiles are used to form an $m \times n$ retangular grid, which they call *mosaics*. Hong and Oh provided bounds on the number of *polygon mosaics*, which are mosaics where the tile patterns form polygons and all tiles that are not part of a polygon are the blank tile. We exactly count polygon mosaics. We also introduce and count mosaics with the related property of not containing a polygon, which we call polygon free mosaics.

1 Mosaics

Imagine you are tasked with tiling a rectangular bathroom floor that is m units by n units, where m and n are positive integers. At your disposal is an unending supply of 7 distinct types of tiles. These tile types, diagrammed in Figure 1, are composed of unit squares with dotted lines connecting 2 sides at their midpoints, as well as the "blank tile".

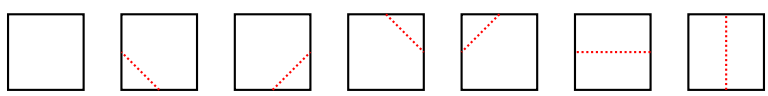


Figure 1: The tile set

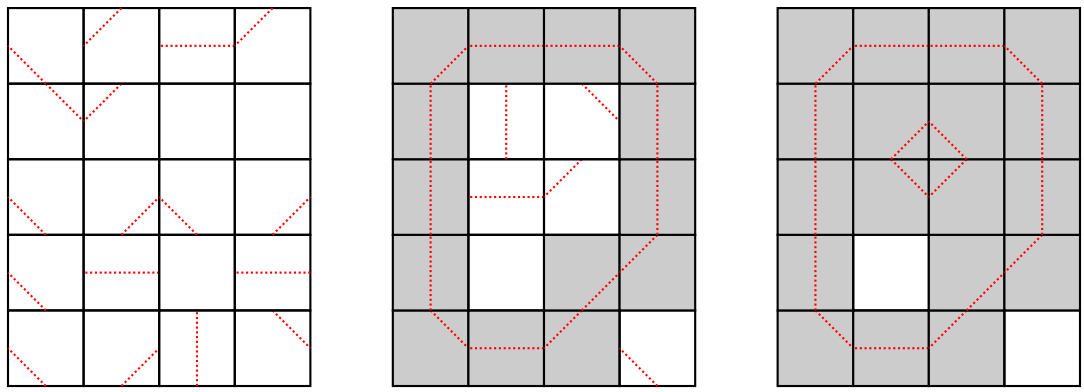
The task is complete once you place mn tiles to cover the floor. We are interested in the number of floors that do not contain polygons[1]. We call a fully tiled $m \times n$ floor an (m, n) *mosaic*. Figure 2 shows various $(5, 4)$ mosaics. Here we highlight in gray the tiles that are part of a polygon.

Definition 1.1. An (m, n) *polygon free mosaic* is an (m, n) mosaic that does not contain a polygon. An (m, n) *polygon mosaic* is an (m, n) mosaic such that any tile that is not part of a polygon is the blank tile.

Figure 2a depicts a polygon free mosaic, Figure 2b depicts a mosaic that is not polygon free, and Figure 2c depicts a polygon mosaic. Notice that the only mosaic that is both a polygon mosaic and a polygon free mosaic is the mosaic of all blank tiles.

To determine the number of polygon free mosaics, we begin by defining a related object to mosaics, that being binary lattices.

[1]Polygons are more commonly referred to as "self-avoiding polygons" in the literature [2, 3] to highlight their connection with self-avoiding walks.



(a) A polygon free mosaic (b) A mosaic that is not polygon free (c) A polygon mosaic

Figure 2: Examples of $(5, 4)$ mosaics

2 Binary Lattices

For positive integers m and n, an (m, n) *binary lattice* is a rectangular lattice of $m + 1$ by $n+1$ vertices, with each vertex labeled 0 or 1. We also define a *framed lattice* to be a binary lattice in which all boundary vertices are labeled 0. An example of a $(5, 4)$ framed lattice is shown on the right of Figure 3. Let a *cell* be a $(1, 1)$ binary lattice. The sixteen possible cells are:

$$\begin{smallmatrix}0&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\1&0\end{smallmatrix},\ \text{and}\ \begin{smallmatrix}1&1\\1&1\end{smallmatrix}.$$

A *proper lattice* is a framed lattice that does not contain any $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cells.

Let $\mathcal{M}$ be a polygon mosaic. We will create a binary lattice from $\mathcal{M}$ by replacing every point that is at a corner of a tile of $\mathcal{M}$ by either 0 or 1. For each such point, we count the number of polygons of $\mathcal{M}$ that surround it, i.e., polygons for which the point is in the polygon's interior. If this number is even, label the vertex 0, and if it is odd, label the vertex 1. We then delete all dotted lines that are part of polygons of $\mathcal{M}$. We define $\mathscr{L}(\mathcal{M})$ to be the resulting binary lattice. See Figure 3 for an example. If $\mathcal{M}$ is an (m, n) polygon mosaic, i.e., m tiles high and n tiles wide, then $\mathscr{L}(\mathcal{M})$ is an (m, n) binary lattice, i.e., $m+1$ vertices high and $n+1$ vertices wide. In fact, $\mathscr{L}(\mathcal{M})$ is an (m, n) framed lattice because every point on the edge of $\mathcal{M}$ is surrounded by no polygons.

In our definition of $\mathscr{L}$, a tile of a polygon mosaic $\mathcal{M}$ corresponds to a cell of $\mathscr{L}(\mathcal{M})$. For a tile of $\mathcal{M}$ that is not the blank tile, consider laying its corresponding cell over it. The dotted line of the tile separates the vertices of the cells, with vertices labeled 0 on one side and labeled 1 on the other. For example, the second from left tile of Figure 1 must map to either the cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ or the cell $\begin{smallmatrix}1&1\\0&1\end{smallmatrix}$. Which one? Consider first where the interior of this polygon is relative to this dotted line, and second, the number of other polygons that contain this polygon. For example, the tile maps to $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ if the interior is below the dotted line and an even number of polygons contain this polygon. On the other hand, a blank tile of $\mathcal{M}$

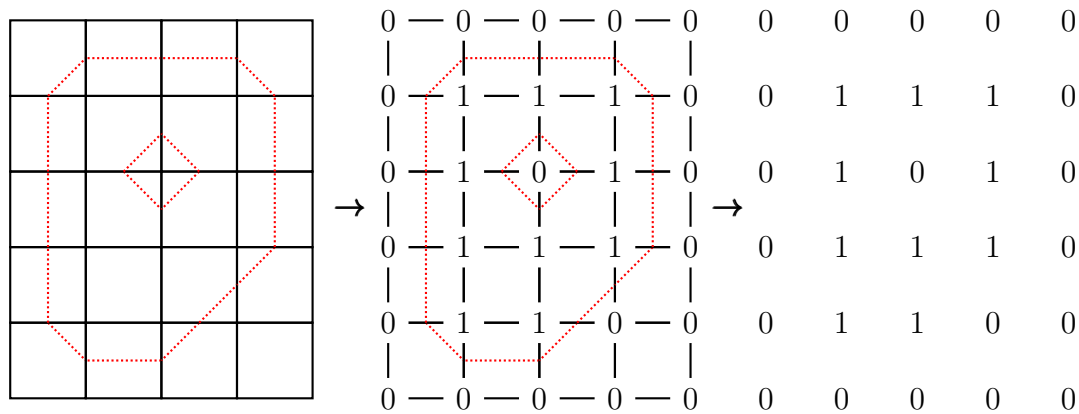


Figure 3: Applying $\mathscr{L}$ to a polygon mosaic

corresponds to a cell that is either $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ or $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$. With seven tiles, and no cell able to be the image of distinct tiles, $\mathscr{L}(\mathcal{M})$ can contain fourteen types of cells. The two types that never appear in $\mathscr{L}(\mathcal{M})$ are $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$. That is, $\mathscr{L}(\mathcal{M})$ is proper.

Given any proper lattice ℓ, we can reverse this process, creating a mosaic by replacing each cell by the unique tile that can map to it. Define $\mathscr{L}^{-1}$ by restricting this map to proper lattices, and let us show that $\mathscr{L}^{-1}$ is indeed an inverse to $\mathscr{L}$, thus proving that $\mathscr{L}$ gives a bijection from (m,n) polygon mosaics to (m,n) proper lattices. Our discussion above shows that $\mathscr{L}^{-1} \circ \mathscr{L}(\mathcal{M}) = \mathcal{M}$ for any polygon mosaic $\mathcal{M}$, so fixing a proper lattice ℓ, we must show that $\mathscr{L}^{-1}(\ell)$ is a polygon mosaic and that $\mathscr{L} \circ \mathscr{L}^{-1}(\ell) = \ell$.

First observe that polygon mosaics can be characterized as those mosaics in which every end of a dotted line on a tile meets, on an adjacent tile, another end of a dotted line. In the mosaic $\mathscr{L}^{-1}(\ell)$, these ends are exactly the points which, in the corresponding cell of ℓ, lie halfway between adjacent vertices of ℓ, one labeled 0 and the other labeled 1. Because ℓ is a framed lattice, the vertex labeled 1 is an internal vertex. Thus, another cell of ℓ shares these two vertices, and the dotted lines of $\mathscr{L}^{-1}(\ell)$ from the two corresponding tiles meet. Therefore $\mathscr{L}^{-1}(\ell)$ is a polygon mosaic.

By our analysis of $\mathscr{L}$ and our definition of $\mathscr{L}^{-1}$, each cell of $\mathscr{L} \circ \mathscr{L}^{-1}(\ell)$ must either be equal to the corresponding cell of ℓ or is formed from the corresponding cell by relabeling all vertices 0 as 1 and 1 as 0. Furthermore, since the cells are connected, i.e., none is isolated, we must then have that $\mathscr{L} \circ \mathscr{L}^{-1}(\ell)$ itself is either equal to ℓ or formed by relabeling all vertices 0 as 1 and 1 as 0. By hypothesis, ℓ is a framed lattice, and by construction, every binary lattice in the range of $\mathscr{L}$ is a framed lattice. Therefore $\mathscr{L} \circ \mathscr{L}^{-1}(\ell) = \ell$. This concludes our proof that $\mathscr{L}^{-1}$ is the inverse of $\mathscr{L}$ and thus that $\mathscr{L}$ is bijective.

The goals of this paper are twofold, counting polygon mosaics and polygon free mosaics.

3 Counting Polygon Mosaics

To count polygon mosaics, we begin by defining a function v that maps cells to real numbers. More generally, we define v from binary lattices to real numbers by taking the product of v

applied to each cell in the binary lattice. Throughout this work we will show that choices of v allow us to count both polygon mosaics (Section 3) and polygon free mosaics (Section 4) under the tile set, as well as certain extensions to the tileset (Section 5).

Consider the v in which

$$\begin{cases} 0 & \text{for cells } \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) \\ 1 & \text{otherwise.} \end{cases}$$

We quickly show that the number of (m, n) polygon mosaics is then $\sum v(\ell)$, where the sum is over all framed lattices ℓ. For the framed lattices that are proper, there is one corresponding polygon mosaic. For the framed lattices that are not proper, the lattice must contain a $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ or $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ cell, and so contribute nothing to the sum.

We can more efficiently compute the sum $\sum v(\ell)$ using matrices for any choice of v. Here we make no choice for the values of v, and so represent specific values of v symbolically. Define the 1×1 matices $A_1 = \left(v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right), B_1 = \left(v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)\right), C_1 = \left(v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)\right), D_1 = \left(v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)\right)$, and for positive integers n, define the following $2^{n-1} \times 2^{n-1}$ matrices

$$A_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)D_n \end{pmatrix} \quad B_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 0 & 0 \\ 1 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)D_n \end{pmatrix}$$

$$C_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)D_n \end{pmatrix} \quad D_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)A_n & v\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)B_n \\ v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)C_n & v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)D_n \end{pmatrix}.$$

Let $\beta_n(k)$ be the n digit binary representation of the nonnegative integer $k < 2^n$, with $\beta_0(0)$ being the empty string.

We extend our shorthand for cells to $(1, n)$ binary lattice using analagous notation and the definition for $\beta_n(k)$. For example, we write

$$v\begin{pmatrix} 0 & \beta_2(1) & 0 \\ 0 & \beta_2(0) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} v\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} v\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

For (m, n) matrices we number the rows $0, 1, \ldots, m-1$ top to bottom, and columns $0, 1, \ldots, n-1$ left to right.

Proposition 3.1. *For positive integer n and nonnegative integers $i, j < 2^{n-1}$, the (i, j)-th entry of*

$$A_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad B_n \text{ is } v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix},$$

$$C_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}, \quad D_n \text{ is } v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix}.$$

Proof. We induct on n. The $n = 1$ case is trivial, as $\beta_0(0)$ is the empty string. We next assume our result for some fixed $n \geq 1$. The induction step requires proving the proposition for entries in $A_{n+1}, B_{n+1}, C_{n+1}$, and D_{n+1}. As each $n+1$ indexed matrix is defined by quadrants of the matrices A_n, B_n, C_n, and D_n, there are sixteen cases to examine. We do two of the sixteen cases for illustration, that being the upper right quadrant of A_{n+1} then the lower left quadrant of D_{n+1}.

Let i, j be nonnegative integers less than 2^{n-1} that index entries in A_n, B_n, C_n, or D_n. For the upper right block matrix in A_{n+1}, by definition the $(i, 2^{n-1}+j)$ entry of A_{n+1} equals $v\left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ times the (i, j) entry of B_n. By our induction hypothesis, this equals

$$v\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & 0 & \beta_{n-1}(i) & 0 \\ 0 & 1 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & \beta_n(i) & 0 \\ 0 & \beta_n(2^{n-1}+j) & 0 \end{pmatrix},$$

as desired. Similarly, for the lower left quadrant of D_{n+1}, by definition the $(2^{n-1}+i, j)$ entry of D_{n+1} equals $v\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ times the (i, j) entry of C_n. By our induction hypothesis, this equals

$$v\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} v\begin{pmatrix} 1 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 1 & 1 & \beta_{n-1}(i) & 0 \\ 1 & 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 1 & \beta_n(2^{n-1}+i) & 0 \\ 1 & \beta_n(j) & 0 \end{pmatrix},$$

again as desired. These arguments must be adapted to prove the desired results for all quadrants of $A_{n+1}, B_{n+1}, C_{n+1}$, and D_{n+1}, but all the work is similar. □

Theorem 3.2. *For all positive integers m and n, the $(0, 0)$ entry of A_n^m equals $\sum v(\ell)$, where the sum is over all (m, n) framed lattices ℓ.*

Proof. Fix n. We first introduce some terminology. For all positive integers m, we say a *height m strip* is an (m, n) binary lattice in which all vertices in the farthest left and farthest right columns are labeled zero. Then for a height m strip ℓ and a nonnegative integer $i < 2^{n-1}$, we say a particular row of ℓ is *determined by i* if that row is equal to $0\ \beta_{n-1}(i)\ 0$. Note that an (m, n) framed lattice is exactly a height m strip in which both the top and bottom rows are determined by 0. Thus, to prove our proposition it suffices to show the following.

Claim. For all positive integers m and nonnegative integers $i, j < 2^{n-1}$, the (i, j) entry of A_n^m equals $\sum v(\ell)$, where the sum is over all height m strips ℓ whose first row is determined by i and whose last row is determined by j.

To prove the claim, first observe that for each i and j, there exists a unique height 1 strip whose first row is determined by i and whose last row is determined by j. By Proposition 3.1, v applied to this strip is equal to the (i, j) entry of A_n. This proves our claim when $m = 1$. We continue by induction, so for some $m \geq 2$, assume the claim for $m - 1$.

Fix nonnegative integers $i, j < 2^{n-1}$, and consider some nonnegative integer $k < 2^{n-1}$. As in the $m = 1$ case, Proposition 3.1 tells us that the (i, k) entry of A_n equals v applied to the unique height 1 strip with first and last rows determined respectively by i and by k. By our induction hypothesis, the (k, j) entry of A_n^{m-1} equals $\sum v(\ell)$, where the sum is over all height $m-1$ strips ℓ whose first and last rows are determined respectively by k and by j. Consider applying v both to the height 1 strip indicated above and to one of the height $m-1$ strips ℓ in this sum. By the definition of v, the product of these two numbers is equal to v applied to the height m strip formed by overlapping the bottom row of the height 1 strip to the top row of ℓ.

$$v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \end{pmatrix} v\begin{pmatrix} 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}$$

From this we see that the product of the (i, k) entry of A_n and the (k, j) entry of A_n^{m-1} is equal to $\sum v(\ell)$, where the sum is now over all height m strips ℓ whose first, second, and

last rows are determined respectively by i, by k, and by j. This tells us, in turn, that when we additionally sum over all nonnegative $k < 2^{n-1}$, we can conclude that the (i, j) entry of $A_n \cdot A_n^{m-1} = A_n^m$ is, as desired, the sum $\sum v(\ell)$ where the sum is over all height m strips ℓ whose first and last rows are determined respectively by i and by j. $\square$

Substituting the values for v from Remark **??** into the matrices from Theorem 3.2 gives the following theorem.

Theorem 3.3. *For positive integers n let A_n, B_n, C_n, D_n be $2^{n-1} \times 2^{n-1}$ matrices with integer entries, where $A_1 = B_1 = C_1 = D_1 = \begin{pmatrix} 1 \end{pmatrix}$ and*

$$A_{n+1} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} \quad B_{n+1} = \begin{pmatrix} A_n & B_n \\ 0C_n & D_n \end{pmatrix}$$
$$C_{n+1} = \begin{pmatrix} A_n & 0B_n \\ C_n & D_n \end{pmatrix} \quad D_{n+1} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}.$$

Then the number of (m, n) polygon mosaics is the $(0, 0)$ entry of A_n^m.

Now that we have counted polygon mosaics, we move to the more complicated polygon free mosaics.

4 Counting Polygon Free Mosaics

We use the following choice for v to count polygon free mosaics

$$\begin{cases} 7 & \text{for cells } \begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix} \\ 0 & \text{for cells } \begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix} \\ -1 & \text{for cells } \begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}, \begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix} \\ 1 & \text{otherwise.} \end{cases}$$

For example, if we let ℓ be the framed lattice on the right of Figure 3, we have $v(\ell) = -7^{13}$, as there are two $\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}$ cells, one $\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}$ cell, twelve $\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}$ cells, and one $\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}$ cell.

We show our choice of v has useful properties by first considering the magnitude, then its sign. For the magnitude of v, for fixed m and n give an index $1 \leq i \leq p$ to every way one polygon can be formed in an (m, n) mosaic. A polygon is determined not only by its shape and orientation, but also its location within the mosaic. For example, Figure 4 depicts three distinct polygons in a $(2, 4)$ mosaic, which for reference we call polygons $1, 2$, and 3.

Let M_i be the set of (m, n) mosaics that contain polygon i. We immediately have the number of mosaics that contain at least one polygon is

$$\left| \bigcup_{i=1}^{p} M_i \right|. \tag{1}$$

Notice that many subsets of polygons do not have mosaics that contains them all. For example, for the polygons in Figure 4, $M_1 \cap M_2 = M_2 \cap M_3 = \varnothing$ but $M_1 \cap M_3$ contains exactly the polygon mosaic with polygons 1 and 3.

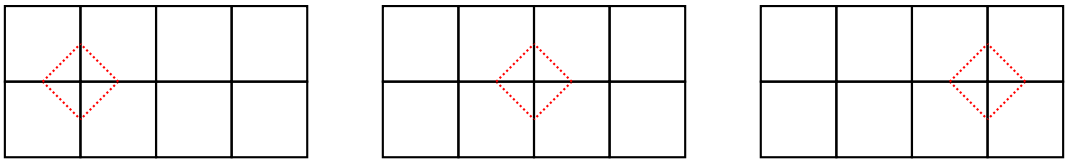


Figure 4: Polygons $1, 2$, and 3

Proposition 4.1. *For a polygon mosaic P, let $\ell = \mathscr{L}(P)$. If J is the subset of indices of the polygons in P we have*

$$\Big|\bigcap_{j\in J} M_j\Big| = \Big|v(\ell)\Big|. \tag{2}$$

Proof. From the definition of M_j, the left hand side of Equation 2 counts the number of mosaics that necessarily contain the polygons in J, and possibly contain other polygons, or tiles of any type that are not part of polygons. For the right hand side, if a cell comes from a tile that is part of a polygon, the absolute value of v of the cell is 1. Therefore, $|v(\ell)|$ is just 7 raised to the number of $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cells in ℓ. As all tiles in the tile set can map to $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$, $|v(\ell)|$ counts all possible tile assignments in all locations that are not part of a polygon in J. Therefore $|v(\ell)|$ counts mosaics that contain the polygons in J. This is the definition of $\bigcap_{j\in J} M_i$, and so the equality holds. □

Proposition 4.2. *For a polygon mosaic P, let $\ell = \mathscr{L}(P)$. If P contains an even [resp. odd] number of polygons then $v(\ell)$ is positive [resp. negative].*

We might want to think of this result as relating a "global" characteristic and "local" characteristic, since the number of polygons in P is determined by looking at the entire mosaic, whereas $v(\ell)$ is calculated by looking at one cell at a time. The reader may wish to compare this to the definition of $\mathscr{L}$, which is done "globally" and the definition of $\mathscr{L}^{-1}$, which was defined "locally."

Proof. For the proof of Proposition 4.2, let N be the number of vertices of ℓ that are 1, and define ℓ_0 to be the lattice formed by changing these N vertices from 1 to 0. That is, ℓ_0 has the same dimensions as ℓ but all vertices are 0. We will change ℓ_0 back to ℓ by flipping these vertices back from 0 to 1, one vertex at a time. The order in which we flip vertices is as follows.

We flip vertices one row at a time, working from top to bottom. In each row we make two passes, each time moving from left to right, and each time flipping a vertex if the corresponding vertex of ℓ is 1. On the first pass we flip a vertex if the vertex is directly above it is 1. (Because we are working the rows from top to bottom, this vertex is 1 in both our current lattice and also in ℓ.) On the second pass we flip the remaining vertices of the row, i.e., if the vertex is directly above it is 0.

This procedure creates a sequence $\ell_0, \ell_1, \ldots, \ell_N$ of lattices with $\ell_N = \ell$. Because $\ell = \mathscr{L}(P)$, we know ℓ is a proper lattice. Since every ℓ_i has the same dimensions as ℓ and has vertices of 0 in at least the same positions as in ℓ, we see every ℓ_i is a framed lattice.

We will prove that every ℓ_i is also a proper lattice (which is the reason we make two passes through each row) and that by taking $P_i = \mathscr{L}^{-1}(\ell_i)$, the proposition holds for P_i and ℓ_i.

$$\begin{array}{ccccc} ul & — & u & — & ur \\ | & & | & & | \\ l & — & \# & — & r \\ | & & | & & | \\ dl & — & d & — & dr \end{array}$$

Figure 5: Cases Diagram

Because every ℓ_i is framed, in the transition from ℓ_{i-1} to ℓ_i, the single vertex that changes is adjacent to eight vertices, as in Figure 5, where the changing vertex # is 0 in ℓ_{i-1} and 1 in ℓ_i, and the remaining vertices are labeled for their relative position to this central vertex, such as dl for down left and u for up.

We need to consider all possible adjacent vertices. Since our process goes from top to bottom one row at a time, we have $(dl, d, dr) = (0, 0, 0)$. Also, because of the way in which we make our two passes, if $u = 1$ then $r = 0$ and $(ul, l) \neq (0, 1)$, whereas if $u = 0$ then $(ur, r) \neq (0, 1)$. Additionally, suppose $u = 0$. Then neither (ul, l) nor (ur, r) equals $(1, 0)$. Indeed, if $(ul, l) = (1, 0)$ then since we are making the second pass through the row, we would have a cell $\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}$ in ℓ_i and thus also in $\ell_N = \ell$, contradicting that ℓ is a proper lattice. Similarly, $(ur, r) = (1, 0)$ implies the that ℓ has a cell $\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}$ (since with $ur = 1$, r cannot change in the second pass), again contradicting that ℓ is a proper lattice. Putting all this together, on the first pass, i.e., when $u = 1$, the vertices can only be as in the six cases pictured in Figure 6. Similarly, on the second pass we have $u = 0$ and one of the six cases of Figure 7.

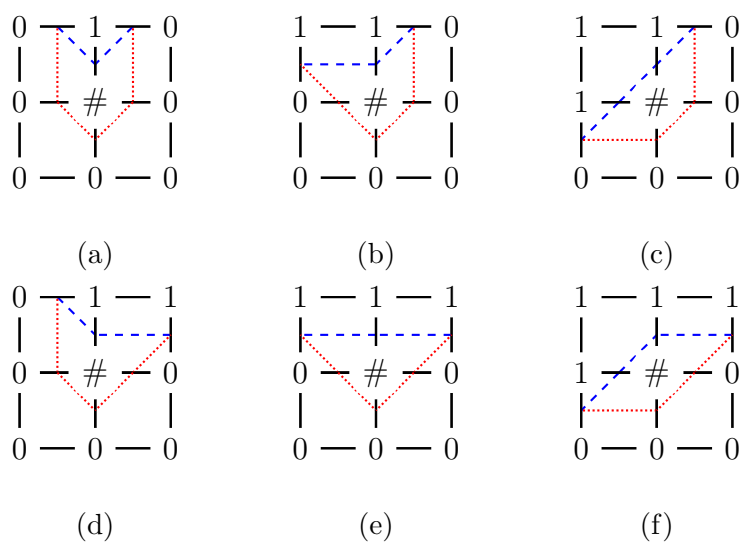


Figure 6: When $u = 1$

Since trivially ℓ_0 is a proper lattice, the twelve cases in Figures 6 and 7 show inductively that every ℓ_i is a proper lattice. For every i, we have already seen that ℓ_i is a framed lattice,

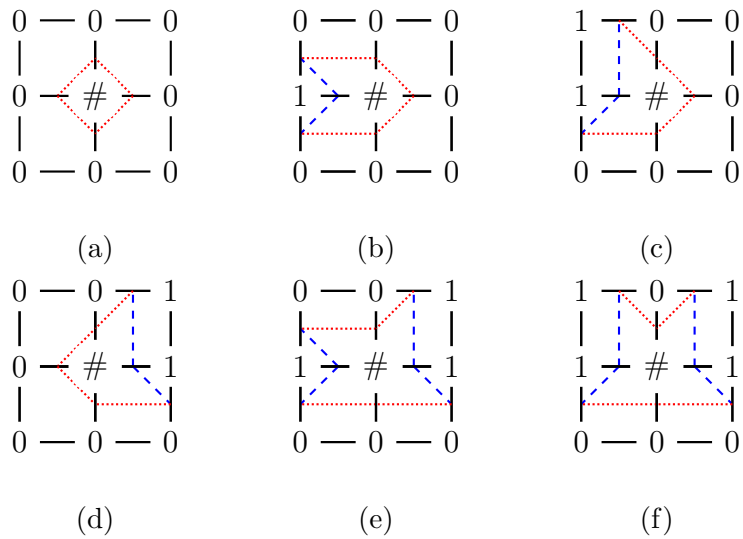


Figure 7: When $u = 0$

so let $P_i = \mathscr{L}^{-1}(\ell_i)$. Figures 6 and 7 also show the portions of P_{i-1} and P_i that are in these two cells, where the dashed lines are from the polygons of P_{i-1} and the dotted lines from the polygons of P_i. (Recall that # is 0 in ℓ_{i-1} and 1 in ℓ_i.)

Let us say a cell is *negative* if it is $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$, i.e., if its evaluation under v is negative. The conclusion of Proposition 4.2 can be rephrased by saying that the number of negative cells in ℓ and the number of polygons in P have the same parity, i.e., are both even or both odd. For example, ℓ_0 satisfies the proposition because ℓ_0 has no negative cells and P_0 has no polygons.

Assume now that some P_{i-1} and ℓ_{i-1} satisfies the proposition, and let us show that the same is true for P_i and ℓ_i. This will complete the proposition. We know that we are in one of the twelve cases of Figures 6 and 7, and we must look at the number of negative cells in ℓ_i versus ℓ_{i-1}, and the number of polygons in P_i versus P_{i-1}. An easy case is (a) of Figure 7 because ℓ_i has one more negative cell (specifically, a cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$), and P_i has one more polygon than P_{i-1}.

Let us now look in the remaining cases at how the number of negative cells changes from ℓ_{i-1} to ℓ_i. In the six cases of Figure 6 and also case (d) of Figure 7, the four pictured cells have no negative cells regardless of whether # equals 0 or 1, so there is no change. The number is also unchanged in (b) of Figure 7 because a single negative cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ changes position between ℓ_{i-1} and ℓ_i. In (c) of Figure 7, the number of negative cells increases by 2 because changing 0 to 1 in the central position creates both a cell $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ and a cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$. Finally, the number decreases by 1 in (e) of Figure 7 and increases by 1 in (f) of Figure 7. Therefore, we can complete the proposition by showing that in the six cases of Figure 6 and in (b), (c), (d) of Figure 7, P_{i-1} and P_i have the same number of polygons, whereas in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1.

Consider, for example, (c) of Figure 6, and look just above the lower left corner and just left of the upper right corner. In P_{i-1} these are connected by a straight dashed line, and in P_i these are connected by dotted line segments. We call both of these *inside connections*.

Because P_{i-1} and P_i are polygon mosaics, these same points are connected by straight line segments outside the pictured area, and we will call these *outside connections* because they connect points on the edge of the pictured square with a path outside the square. This outside connection is the same in P_{i-1} and P_i. Therefore P_{i-1} and P_i have the same number of polygons because the polygon with this outside connection changes shape, and all other polygons are unchanged. Similar reasoning completes the remaining cases of Figure 6 and cases (b), (c), (d) of Figure 7.

We now must show that in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1. We will explicitly show this in (e), and essentially identical reasoning applies to (f). There are four places where inside connections meet the edge of the outer square. Beginning in the lower left and working clockwise, let x_1, x_2, x_3, and x_4 be these points. For example, x_3 is just left of the upper right corner and x_4 is just above the lower right corner. In P_{i-1}, there is an inside connection between x_1 and x_2, and another between x_3 and x_4, whereas P_i has inside connections between x_1 and x_4, and between x_2 and x_3.

Any outside connections in P_{i-1} and P_i must be identical, and since these are polygon mosaics, each of x_1, x_2, x_3, and x_4 is the end of some outside connection. One possibility is that we have one outside connection between x_1 and x_2, and another between x_3 and x_4. In this case, the inside and outside connections of P_{i-1} complete two polygons, one including x_1 and x_2, and the other including x_3 and x_4. However, P_i will have one fewer polygon because a single polygon contains all x_i. Another possibility is to have one outside connection between x_1 and x_4, and another between x_2 and x_3. In this case, in P_{i-1} a single polygon includes all x_i, whereas in P_i they complete two polygons, one containing x_1 and x_4, and the other containing x_2 and x_3, and therefore P_i has one more polygon than P_{i-1}. These are the only possibilities since it is impossible to have an outside connection between x_1 and x_3, and another between x_2 and x_4. This is because our polygons are in a plane, and polygons in a mosaic are not able to cross one another. We have completed our proof. □

Theorem 4.3. *For all positive integers m and n, the number of (m,n) polygon free mosaics is $\sum v(\ell)$, where the sum is over all (m,n) framed lattices.*

Proof. From Equation 1 we have the number of polygon free mosaics is

$$7^{mn} - \Big|\bigcup_{i=1}^{p} M_i\Big|.$$

By the inclusion-exclusion principle (see the Appendix), we have

$$-\Big|\bigcup_{i=1}^{p} M_i\Big| = \sum_{\varnothing \neq J \subseteq \{1,2,\ldots,p\}} (-1)^{|J|}\Big|\bigcap_{j\in J} M_j\Big|. \tag{3}$$

For Equation 3, for some $\varnothing \neq J \subseteq \{1,2,\ldots,p\}$, if $\bigcap_{j\in J} M_j$ is empty then the $\Big|\bigcap_{j\in J} M_j\Big|$ term is 0. Therefore, the domain of summation can be rewritten to be over all J such that there exists a polygon mosaic P with exactly the polygons in J. For fixed J, if $\mathscr{L}(P) = \ell$, Propositions 4.1 and 4.2 give the following

$$(-1)^{|J|}\Big|\bigcap_{j\in J} M_j\Big| = v(\ell).$$

Therefore, we can rewrite the sum in Equation 3 as

$$\sum_{J\subseteq\{1,2,\ldots,p\}} (-1)^{|J|}\Big|\bigcap_{j\in J} M_j\Big| = \sum_{\ell} v(\ell).$$

where ℓ is over all proper lattices that $\mathscr{L}^{-1}(\ell)$ has at least one polygon. Additionally notice that v applied to the binary lattice made up of all $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ cells equals 7^{mn}, and is the only proper lattice such that $\mathscr{L}^{-1}(\ell)$ has no polygons. This gives

$$7^{mn} - \Big|\bigcup_{i=1}^{p} M_i\Big| = \sum_{\ell} v(\ell),$$

where the sum is over all proper lattices. Finally, notice that the sum $\sum_\ell v(\ell) = 0$ for the framed lattices that contains $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$, as $v\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right) = v\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right) = 0$. Therefore, summing over all framed lattices in Equation **??** gives the desired result. □

Theorem 4.3 let's us apply Theorem 3.2 to the choice of v at the beginning of this section, which gives the main result of the paper.

Theorem 4.4. *For positive integers n let A_n, B_n, C_n, D_n be $2^{n-1}\times 2^{n-1}$ matrices with integer entries, where $A_1 = \begin{pmatrix}7\end{pmatrix}, B_1 = \begin{pmatrix}-1\end{pmatrix}, C_1 = \begin{pmatrix}1\end{pmatrix}, D_1 = \begin{pmatrix}1\end{pmatrix}$ and*

$$A_{n+1} = \begin{pmatrix}7A_n & B_n\\ C_n & D_n\end{pmatrix} \quad B_{n+1} = \begin{pmatrix}-A_n & B_n\\ 0C_n & D_n\end{pmatrix}$$
$$C_{n+1} = \begin{pmatrix}A_n & 0B_n\\ C_n & D_n\end{pmatrix} \quad D_{n+1} = \begin{pmatrix}A_n & -B_n\\ C_n & 7D_n\end{pmatrix}.$$

Then the number of (m,n) polygon free mosaics is the $(0,0)$ entry of A_n^m.

Finally, we introduce possible extensions to our work, as well as how our work relates to other mosaic counting efforts.

5 Related Work and Extensions

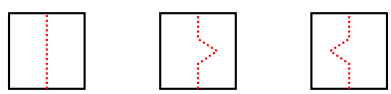
In terms of related work, our work can be seen as an extension to Hong and Oh [4] and further generalizing the techniques in Oh et al. [11] to explore a new direction in counting mosaics. Hong and Oh [4] provided bounds on the number of polygon mosaics.

Theorem 5.1 ([4])**.** *The number of (m,n) polygon mosaics for $m,n\geq 3$ is bounded between $2^{m+n-3}\left(\frac{17}{10}\right)^{(m-2)(n-2)}$ and $2^{m+n-3}\left(\frac{31}{16}\right)^{(m-2)(n-2)}$.*

Our Theorem **??** compliments this result with a procedure to exactly count polygon mosaics, as well as exactly recreate the sequence A181245 on the OEIS [6]. Note that unlike here, the authors in [4] did not consider the mosaic made from only blank tiles a polygon mosaic, and so our result differs to theirs by 1. We also point out that our work allows for both polygon and polygon free mosaics to be counted. The methods in [11] could be used to count just polygon mosaics (i.e. an analagous result to Theorem **??**), a fact which is likely known to the authors of [4] and [11], though to our knowledge has not appeared in print.

In related work, Lomonaco and Kauffman [5] introduced mosaics constructed from a tile set of 11 distinct tiles, of which our tile set is a subset. The authors were interested in a subset of mosaics which they call *knot mosaics*. Oh et al. [11] counted the number of knot mosaics using an analagous method to that of Section **??**. Oh and colleagues go beyond counting by bounding the growth rate of knot mosaics [8, 10, 1], and Oh further adapts techniques similar to Section **??** to solve problems in monomer and dimer tilings [7, 9]. Related ideas were independently used to count the number of rectangular partitions of a rectangle in [12] for the sequence A182275 on the OEIS [6].

Counting the number of polygon free mosaics in certain extensions of the tile set is simple to do. For example, let's extend our tile set to include three types of tiles that connect the top and bottom midpoints like below:



To count polygon [resp. polygon free] mosaics in this new tile set, one simply needs to adjust the choice of v in Section 3 [resp. 4] to have $v\left(\begin{smallmatrix}0&1\\0&1\end{smallmatrix}\right) = v\left(\begin{smallmatrix}1&0\\1&0\end{smallmatrix}\right) = 3$.

We point out it is not possible to extend our results naively to the Lomonaco and Kauffman [5] tile set to count the number of *knot free mosaics*. This is specifically due to the reasoning for Case 7e and Case 7f in the proof of Theorem 4.2, as knots are able to cross each other.

6 Acknowledgements

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References

[1] Dooho Choi et al. "Quantum knot mosaics and bounds of the growth constant". In: *Reviews in Mathematical Physics* 36.10 (2024), p. 2450025. DOI: 10.1142/S0129055X24500259. eprint: https://doi.org/10.1142/S0129055X24500259. URL: https://doi.org/10.1142/S0129055X24500259.

[2] Paul J Flory. "The configuration of real polymer chains". In: *The Journal of Chemical Physics* 17.3 (1949), pp. 303–310.

[3] Anthony John Guttmann. "Polygons, Polyominoes and Polycubes". In: *arXiv: Combinatorics* (2009), pp. 43–78. URL: https://api.semanticscholar.org/CorpusID:122240634.

[4] Kyungpyo Hong and Seungsang Oh. "Bounds on Multiple Self-avoiding Polygons". In: *Canadian Mathematical Bulletin* 61.3 (Sept. 2018), pp. 518–530. ISSN: 1496-4287. DOI: 10.4153/cmb-2017-072-x. URL: http://dx.doi.org/10.4153/CMB-2017-072-x.

[5] Samuel J. Lomonaco and Louis H. Kauffman. "Quantum knots and mosaics". In: *Quantum Information Processing* 7.2 (2008), pp. 85–115. DOI: 10.1007/s11128-008-0076-7. URL: https://doi.org/10.1007/s11128-008-0076-7.

[6] OEIS Foundation Inc. *The On-Line Encyclopedia of Integer Sequences*. Published electronically at http://oeis.org.

[7] Seungsang Oh. "Domino tilings of the expanded Aztec diamond". In: *DISCRETE MATHEMATICS* 341.4 (Apr. 2018), pp. 1185–1191. ISSN: 0012-365X. DOI: 10.1016/j.disc.2017.10.016.

[8] Seungsang Oh. "Quantum knot mosaics and the growth constant". In: *Topology and its Applications* 210 (2016), pp. 311–316. ISSN: 0166-8641. DOI: https://doi.org/10.1016/j.topol.2016.08.011. URL: https://www.sciencedirect.com/science/article/pii/S0166864116301857.

[9] Seungsang Oh. "State matrix recursion method and monomer-dimer problem". In: *DISCRETE MATHEMATICS* 342.5 (May 2019), pp. 1434–1445. ISSN: 0012-365X. DOI: 10.1016/j.disc.2019.01.022.

[10] Seungsang Oh and Youngin Kim. "Growth rate of quantum knot mosaics". In: *Quantum Information Processing* 18.8 (2019), p. 238. DOI: 10.1007/s11128-019-2353-z. URL: https://doi.org/10.1007/s11128-019-2353-z.

[11] Seungsang Oh et al. "Quantum knots and the number of knot mosaics". In: *Quantum Information Processing* 14.3 (2015), pp. 801–811. DOI: 10.1007/s11128-014-0895-7. URL: https://doi.org/10.1007/s11128-014-0895-7.

[12] Joshua Smith and Helena Verrill. *On Dividing Rectangles Into Rectangles*. Published electronically at https://oeis.org/A116694/a116694.pdf.

7 Appendix

Proposition: The Inclusion-Exclusion Principle: For finite sets $A_1, A_2, \ldots, A_M$,

$$\left|\bigcup_{j=1}^{M} A_j\right| = \sum_{\emptyset \neq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j \in I} A_j\right|$$

Proof. The $M = 1$ case is trivial, and $M = 2$ case is the familiar and easily justified statement that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. Proceeding inductively, for $M \geq 3$, assume the result for $M - 1$ sets, and let us prove it for M sets. Let us split up the sum on the right into those sets I that do not contain M and those that do. By our inductive hypothesis, the first sum equals $|\bigcup_{j=1}^{M-1} A_j|$ and the second sum can be written as $\sum_{\{M\} \subseteq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j \in I} A_j\right|$. Separating out the term with $I = \{M\}$, this equals $|A_M| - \sum_{\emptyset \neq I \subseteq \{1,2,\ldots,M-1\}} (-1)^{|I|} |\bigcap_{j \in I} (A_j \cap A_M)|$. Again using our inductive hypothesis, the sum here is $\left|\bigcup_{j=1}^{M-1} (A_j \cap A_M)\right| = \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. The entire right side of our desired equation therefore equals $\left|\bigcup_{j=1}^{M-1} A_j\right| + |A_M| - \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. By the $M = 2$ case, this equals $|\bigcup_{j=1}^{M} A_j|$, as desired. $\square$